

House Price Prediction Using Forward & Backward Propagation

Comparative Empirical Analysis of ReLU vs. Tanh Activation Dynamics in Single-Layer Neural Networks

1. EXECUTIVE SUMMARY & INTRODUCTION

This study evaluates a foundational single-layer neural network implemented strictly from first principles (without reliance on high-level framework abstractions like `model.fit()`) to model residential real estate valuations. The core objective is to analyze the exact mathematical mechanics governing feedforward computation, error backpropagation, parameter updates, and gradient behavior.

Specifically, this research investigates the comparative learning dynamics and generalization performance between non-linear **Rectified Linear Unit (ReLU)** and **Hyperbolic Tangent (Tanh)** activation functions under identical structural and optimization hyperparameters.

DATASET SIZE

3,000

Observations (11 Features)

BEST TEST RMSE

₹29,010

Achieved via ReLU Model

OPTIMIZATION

2,000

Epochs @ Learning Rate 0.01

2. DATASET TOPOLOGY & MISSING VALUE PROFILE

The underlying corpus contains 3,000 records tracking key architectural, geographical, and demographic factors. A preliminary audit confirmed zero duplicate records, with missing data isolated across exactly six attributes (75 missing entries each, representing 2.5% per column). The target variable (*House Price*) exhibited complete integrity.

Feature Name	Data Type	Missing Count	Imputation Strategy
Bedrooms, Bathrooms, Sq Footage	Numerical	75 each	Training Set Median
House Age, Distance to Center, Lot Size	Numerical	75 each	Training Set Median
Has Garage	Binary Category	75	Training Set Mode
Neighborhood Quality	Categorical (3 levels)	0	One-Hot Encoded (N-1)
Year Built, Number of Floors	Numerical	0	Standardized (StandardScaler)

3. DATA PREPROCESSING PIPELINE

Train-Test Partitioning: Data was split in an 80:20 ratio (2,400 training samples, 600 test samples) utilizing a fixed seed (`random_state=42`) to isolate evaluation on unseen data.

Encoding & Normalization: *Neighborhood Quality* (Low, Medium, High) was transformed via one-hot encoding, dropping the first category to eliminate multicollinearity. Feature matrices were standard-scaled

4. MODEL ARCHITECTURE & FORWARD PROPAGATION

The network comprises 11 preprocessed input features, 11 trainable weights **W**, 1 bias scalar **b**, and 1 output unit. Parameters were initialized deterministically: **W** ~ Uniform × 0.01, **b** = 0.

Linear Combination

$$z = XW + b$$

Output Hypothesis

$$\hat{y} = \text{Activation}(z)$$

Rectified Linear Unit (ReLU)

$$\text{ReLU}(z) = \max(0, z)$$

Derivative: 1 for $z > 0$, else 0. Passes gradients directly without attenuation for positive activations.

Hyperbolic Tangent (Tanh)

$$\text{Tanh}(z) = \tanh(z)$$

Derivative: $1 - \tanh^2(z)$. Bounded in $[-1, 1]$. Subject to gradient saturation at high magnitude inputs.

5. BACKWARD PROPAGATION & GRADIENT DESCENT

Loss minimization utilized Mean Squared Error (MSE). Optimization was performed via explicit chain rule backpropagation across 2,000 full-batch gradient descent iterations at a fixed learning rate $\eta = 0.01$:

$$L = (1/n) \sum (y - \hat{y})^2 \Rightarrow \partial L / \partial \hat{y} = (2/n)(\hat{y} - y)$$

$$\partial L / \partial z = (\partial L / \partial \hat{y}) \times (\partial \hat{y} / \partial z) \quad | \quad dw = x^T (\partial L / \partial z) \quad | \quad db = \sum (\partial L / \partial z)$$

6. EMPIRICAL PERFORMANCE COMPARISON

Performance metrics across both training and test evaluation phases demonstrate a structural advantage for ReLU within this specific optimization setup:

Activation Function	Final Train MSE (Scaled)	Test Set MSE (Unscaled)	Approximate Test RMSE	Final Gradient Norm
ReLU	0.001871	841,552,139.42	≈ ₹29,010	1.33×10^{-10}
Tanh	0.002406	1,117,663,201.05	≈ ₹33,432	4.70×10^{-6}
Delta (Δ)	+0.000535 (Tanh Higher)	+276,111,061.63	+₹4,422 (+15.2%)	—

7. COMPUTATIONAL EXECUTION FLOW



8. CONVERGENCE & GRADIENT DYNAMICS ANALYSIS

Loss Curve & Error Trajectory

Both activation profiles demonstrated smooth monotonic error reduction across the initial 500 epochs, confirming appropriate learning rate selection without severe step-size oscillations. However, ReLU established a lower loss floor early and maintained a consistent margin of lower residual error through Epoch 2,000.

Gradient Saturation Mechanics

The gradient norm for Tanh settled at 4.70×10^{-6} compared to ReLU's 1.33×10^{-10} . The higher residual gradient in Tanh combined with higher loss reflects derivative squashing in outer activation regions:

Activation Function → Derivative ($\partial \hat{y} / \partial z$) → Backprop Gradient → Parameter Update → Fit Quality

As pre-activation values z expand, Tanh derivatives approach zero ($1 - \tanh^2(z) \approx 0$), suppressing weight updates even when prediction error remains substantial. In contrast, ReLU yields a constant gradient factor of 1 for all positive inputs, allowing uninterrupted error propagation.

9. ARCHITECTURAL LIMITATIONS & METHODOLOGICAL CONSTRAINTS

Interpretation Scope & Best Practices

The superior outcome of ReLU in this benchmark must not be generalized as a universal assertion that ReLU supersedes Tanh in all neural networks. Performance is tightly coupled to dataset geometry, scaling, and network depth:

- Bounded Target Conflicts:** Single-layer regression typically employs linear output activations. Applying Tanh bounds predictions strictly between $[-1, 1]$, requiring artificial target scaling.
- Zero-Centered Gradient Flow:** Tanh outputs are zero-centered, which can benefit deeper networks by preventing directional bias in gradient updates—a property absent in basic ReLU units.

10. KEY EVALUATION METRICS SUMMARY

MSE (MEAN SQUARED ERROR)

Penalizes large outlier errors quadratically. Primary loss metric optimized during training.

MAE (MEAN ABSOLUTE ERROR)

Provides direct error measurement in raw currency units (₹) without quadratic distortion.

R² COEFFICIENT

Quantifies proportion of price variance explained relative to baseline mean model.

11. FINAL CONCLUSIONS

This empirical investigation successfully implemented and validated a end-to-end neural network learning pipeline from scratch. The experimental evidence demonstrates that for continuous real estate valuation under single-layer architecture:

- ReLU outperforms Tanh** across test accuracy metrics, reducing RMSE from ₹33,432 to ₹29,010 (a 15.2% error reduction).